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MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR
....
COMMISSION NATIONALE D'ORGANISATION DE L'EXAMEN
HIGHER NATIONAL DIPLOMA EXAM (HND)
....

REPUBLIC OF CAMEROON
Peace-Work-Fatherland
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MINISTRY OF HIGHER EDUCATION
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NATIONAL COMMISSION FOR THE ORGANISATION OF
HIGHER NATIONAL DIPLOMA EXAM (HND) EXAM
....

National Exam of Higher National Diploma 2023 Session

Spécialty/option: NWS, SWE, EDM
Paper : DISCRETE MATHEMATICS **Credits : 4/4/10**
Duration : 4 hours

Instructions to candidates: Answer ALL questions

SECTION A: MCQS – (20 marks)

1) A ballot box contains 6 blue balls, 4 red balls and 2 green balls indiscernible on touch. One picks at random and simultaneously picks 3 balls. The probability not to obtain 3 balls of the same colour is:

   a) $\frac{6^3+4^3}{12^3}$
   b) $\frac{49}{55}$
   c) $\frac{6}{55}$
   d) $\frac{6\times5\times4+4\times3}{220}$

2) The code to access a residential building has 4 digits to be dialled on a numeric keypad with 10 digits from 0 to 9. We know that the code has exactly two digits of « 5 ». The number of possible codes is:

   a) 9^2
   b) 2×9^2
   c) 3×9^2
   d) 6×9^2

3) A player throws a biased die having six faces ones. He gains 10 points if 1 appears and 1 point if 2 or 4 appear. He gains no points if other number appears. Given that X is the Radom Variable for the number of points gained by the payer. The variance of X is :

   a) 2
   b) 13
   c) 16
   d) 17

4) The correlation between two quantitative variable X and Y is strong if the coefficient of correlation r is such that:

a) $|r| \geq 0.87$;

b) r approximates to -1;

c) r approximates to 0;

d) r approximates to *1.5*

5) The value of: $\lim_{x\to+\infty} 2e^{-2x}$ is:

a) 0

b))1

c) 2

d) $+\infty$

6) Let $I_n = \int_0^1 x^n e^{-x}\,dx$; where n is natural number. Then. $\lim_{n\to+\infty} I_n$ is equal to:

a) $+\infty$

b) $-\infty$

c) 0

d) 1

7) Let us consider the function $f(x) = \frac{e^{2x}-1}{2e^x}$. Then:

a) $f^{-1}(x) = \ln(x - \sqrt{x^2+1})$

a. $f^{-1}(x) = \ln(x + \sqrt{x^2-1})$

b. $f^{-1}(x) = \ln(x + \sqrt{x^2+1})$

c. $f^{-1}(x) = \ln(x - \sqrt{x^2-1})$

8) Let X1, X2, X3 be uniform random variables on the interval (0, 1) with Cov(Xi, Xj) = 1/24 for i, j =1, 2, 3, i≠j. Calculate the variance of X1 + 2X2 − X3.

a) 1/6

b) 5/12

c) 1/4

d) ½

9) Given that $X \sim B(n, 0.3)$ and E(X) = 4.6, the value of n is

a) 12.80

b) 15.33

c) 4.90

d) 48.00

10) The probability distribution function of a continuous random variable X is

$$f(x) = \begin{cases} kx(2-x), & 0 \leq x \leq 2 \\ 0, & else\ where \end{cases}$$

The value of the constant K is

a) $\frac{4}{3}$

b) $\frac{3}{20}$

c) $\frac{3}{4}$

d) 1

11) Let f be a continuous and strictly decreasing function defined in $[a, b]$, a < b. Let

$f^{-1}(x) = g(x)$. then

a) $g'(x)<0$

b) $g'(x) > 0$

c) $g(x) < 0$ D.

d) $g(x) >0$

12) Given a sequence(U_n) defined by $U_o = 0$ and $\forall$ n ≥ 1, $U_n = \sqrt{U_{n-1} + 6}$, where n is a natural number. The sequence (U_n) is

a) Divergent
b) Convergent
c) Decreasing
d) Constant

13) The integral $I = \int 7^{3x}\, dx$ is equal to:

a) $I = 3(7^{3x})$

b) $I = \frac{7^{3x}}{3}$

c) $I = \frac{7^{3x}\ln 7}{3}$

d) $I = (3\ln 7)7^{3x}$

14) The value of $\sum_{k=1}^{n}(\frac{1}{k}-\frac{1}{k+1})$ is;

a) $\frac{1}{n+1}$

b) $\frac{n}{n+1}$

c) $\frac{n+1}{n}$

d) $\frac{1}{n+1}-1$

15) Let us consider $\boldsymbol{u} = -2\boldsymbol{i} + \boldsymbol{j} + \boldsymbol{k}$, $\boldsymbol{v} = \boldsymbol{i} - \boldsymbol{j} + \boldsymbol{k}$ and $\mathbf{w} = \mathbf{i} - \mathbf{j} + 2\mathbf{k}$. Then:

a) **v.w** = 0

b) **u.v** = 2

c) **w.u** = -1

d) **w.v** =5

16) If $y = (ax+1)e^{2x}$ is a solution to the differential equation $y'' - 2y' - y = (x-1)e^{2x}$

, then $a =$

a) 2

b) $\frac{1}{2}$

c) -1

d) -2

17) The domain of the function $f: \mathbb{R} \longrightarrow \mathbb{R}$ defined by $f(x) = \sqrt{\frac{x-1}{x}}$ is

a) $\{x: x < 0 \text{ and } x \geq 2\}$

b) $\{x: x < 0 \text{ or } x \geq 2\}$

c) $\{x: x < 0\}$

d) $\{x: x \geq 2\}$

18) A and B are two independent events with $p(A) = 0.5$ *and* $p(B) = 0.2$. The probability of the event A∪B is

a) 0.1

b) 0.7

c) 0.6

d) 0.8

19) In a shop, a tray contains marked down notebooks. We know that 50% notebooks are spiral-bound and 75% of notebook are large-squared. Among the large-squared notebooks, 40% are spiral-bound. Larissa randomly chooses a spiral-bound notebook. The probability that it is large squared is

a) 0.3

b) 0.5

c) 0.6

d) 0.75.

20) The function $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 2x^3 + x$

a) A. periodic

b) neither odd nor even

c) even

d) odd

21) For $y = \sin^2 x + \cos^2 x$, $\frac{dy}{dx} =$

a) $2\sin x - 2\cos x$

b) $2\sin x \cos x$

c) $4\sin x \cos x$

d) 0

22) If $y^3 = x^2$, then $\frac{dy}{dx} =$

a) $\frac{2x}{y^3}$

b) $\frac{2x}{3y^2}$

c) $\frac{x^2}{3y^2}$

d) $\frac{x^2}{3y}$

23) $\csc\left(-\frac{4}{3}\pi\right) =$

a) 2

b) $\frac{\sqrt{3}}{2}$

c) $\frac{2\sqrt{3}}{3}$

d) $-\frac{2\sqrt{3}}{3}$

24) Find a positive value c, for x, that satisfies the conclusion of the Mean Value Theorem for Derivatives for $f(x) = 3x^2 - 5x + 1$ on the interval [2,5].

a) 1

b) $\frac{7}{2}$

c) 3

d) $\frac{23}{6}$

25) At what value of x does the function $f(x) = \frac{(x+1)^2}{x^2-1}$ have a removable discontinuity?

a) 1

b) 3

c) 2

d) -1

26) Suppose g is a function defined on the interval [0, 5] such that $\lim_{x \to 2} g(x) = -1$, find

$\lim_{x \to 2} [g(x)]^3$

a) 8

b) $\frac{1}{8}$

c) -1

d) $\frac{1}{3}$

27) If $f(a) = f(b) = 0$ and $f(x)$ is continuous on [a, b], then

a. $f(x)$ must be identically zero

b) $f'(x)$ may be different from zero for all x on (a, b)

c) There exists at least one number c, $a < c < b$, such that $f'(c) = 0$

d) $f'(x)$ must exist for every x on (a, b)

28) A medical treatment has a success rate of .8. Two patients will be treated with this treatment. Assuming the results are independent for the two patients, what is the probability that neither one of them will be successfully cured?

a) 0.5

b) 0.36

c) 0.2

d) 0.04

29) Suppose that the probability of event A is 0.2 and the probability of event B is 0.4. Also, suppose that the two events are independent. Then P(A|B) is:

a) P(A) = 0.2

b) P(A)/P(B) = 0.2/0.4 = ½

c) P(A) × P(B) = (0.2)(0.4) = 0.08

d) None of the above.

30) A specific range of numbers within which a population mean should lie is

a) the range.

b) the confidence coefficient.

c) the confidence interval.

d) the confidence level.

31) Calculate the sample standard deviation using the following data:

X: 4 5 2 6 4 3 4

The standard deviation is which of the following?

a) 10

b) 1.66

c) 1.20

d) 1

33) A sample of 40 cows is drawn to estimate the mean weight of a large herd of cattle. If the standard deviation of the sample is 96 kg, what is the maximum error in a 90% confidence interval estimate?

a) 25 kg

b) 158 kg

c) 58 kg

d) 30 kg

34) The Laplace transform of f(t) = 3 is

a) $\frac{3}{s}$

b) $\frac{1}{s}$

c) 3

d) 3s

34) The inverse Laplace transform of **Erreur !** is

a) 1 – **Erreur !t**

b) 1 – **Erreur !t** + **Erreur !t**

c) 1 – **Erreur !t** + **Erreur !t**^{2}

d) **Erreur !t** + **Erreur !t**^{2}

35) The eigenvalues of $A = (1\ \ 3, 4\ \ 2)$ are.

a) 5 and -2

b) 2 and 5

c) 3 and 1

d) -3 and -1

36) The interval of convergence the function $f(x) = x^2 \ln(1-2x)$

a) $I = (1, 2)$

b) $I = [-\frac{1}{2}, \frac{1}{2})$

c) $I = [1, 2]$

d) $I = [-\frac{1}{2}, \frac{1}{2}]$

37) $f(x)=\sqrt{4+x}$ as a Maclaurin series has as it's radius of convergence

a) R = 2
b) R = 0
c) R = 3
d) R = 1

38) $\sec^2 x - \tan^2 x$ is an identity for which of the following:

a) A. 1
b) $\sin^2 x - \cos^2 x$
c) $\cos^2 x - \sin^2 x$
d) $1 - \tan^2 x$

39) What is the magnitude of the vector $6\mathbf{i} + 15\mathbf{j}$?

a) $\sqrt{21}$
b) 261
c) 21
d) $\sqrt{261}$

40) What are the first five terms of the sequence defined as

$a_1 = 3$

$a_{n+1} = a_n - 4$, for $n \geq 1$?

a) $-3, -2, -1, 0, 1$
b) $-1, -5, -9, -13, -17$
c) $3, -1, -5, -9, -13$
d) $3, -1, 0, 1, 2$

SECTION B: STRUCTURAL

1. **ANALYSIS(30 marks)**

1.1 State Rolle's Theorem. Find the value of x_0 prescribe in Rolle's Theorem for $f(x) = x^3 - 12x$ on the interval $0 \leq x \leq 2\sqrt{3}$. **(5 marks)**

1.2 Evaluate the double integral below. **(5 marks)**

$$\int_{-1}^{2}\int_{2x^2-2}^{x^2+x} x\, dydx.$$

1.3 If $f(x,y) = x^3e^{5y} + y\sin 2x$. Find $f_x, f_y, f_{xx}, f_{yy}, f_{xy}$

(5 marks)

1.4 Find a) $\nabla.\vec{F}$ (div($\vec{F}$)) b) grad (div($\vec{F}$)) and c) $\nabla \times \vec{F}$ (curl($\vec{F}$) if

$\vec{F} = (y + x^2)\vec{\imath} + z\vec{\jmath} + x^2\vec{k}$ (5 marks)

1.5 Use Laplace transform to solve the dy following differential equation:

$\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 10y = e^{2x} + 20$. Given that when $x = 0, y = 0,\ \frac{dy}{dx} = -\frac{1}{3}$ (10 marks)

2. STATISTICS (30 marks)

2.1 An inquiry in the salary of workers of an enterprise is concerned with the sample of 58 workers. The results of the inquiry are shown in the table below:

Hourly Salary(CFA)	1440 - 1620	1620 - 1800	1800 - 1980	1980 - 2160	2160 - 2340	2340 – 2520	2520 - 2700
Number of workers	X	2	4	8	11	15	Y

a) Knowing that the average hourly salary of 58 workers of the enterprise is 2302,76CFA, determine the missing frequency values, x and y. **(9 Marks)**

b) Determine the median salary of the 58 workers of the enterprise. **(4 Marks)**

c) Calculate the first and the third quartile. **(3 Marks)**

2.2 A Credit Union has a portfolio of four branches Edea (E), Garoua (G), Douala(D), Limbe (L).

These branches are expected to generate profits as follows:

Branch	Edea (E)	Garoua (G)	Douala(D)	Limbe (L)
Profit (000frs)	2000	3000	4500	6000

The likelihood that each branch will generate the expected profit is as follows:

Branch	Edea (E)	Garoua (G)	Douala(D)	Limbe (L)
Probability	0.3	0.2	0.1	0.4

a) Calculate the expected profit that will be generated by Credit Union. **(6 Marks)**

b) Determine the variance of the profits. **(6 Marks)**

c) Calculate the standard deviation of profits. **(2 Marks)**

3. PROBABILITY (20 marks)

3.1

a) The probability distribution of a discrete random variable X is given by

$$P(X = x) = \begin{cases} k(x+1), & x = 1,2,3 \\ k(x-3), & x = 4,5,6 \\ 0, & elsewhere \end{cases}$$

Find ;

i. the value of the constant k.

ii. the mean of X

iii. the variance of the distribution **(3+2+4 = 9 Marks)**

3.2 Of the large number of candidates applying for a job in an enterprise, 5% are qualified. The candidates are interviewed one after another until the first qualified applicant is interviewed.

a) Find the mean number of applicants interviewed.

b) Calculate, to 4 decimal places, the probability that the fifth candidate interviewed is the first qualified applicant. **(2+3 = 4 marks)**

3.3 Two machines X and Y produce, respectively, 60 % and 40 % of the total number of electrical fuses from a certain factory. The probabilities of faulty fuses produced by these machines are $\frac{2}{25}$ *and* $\frac{1}{20}$respectively. A fuse is selected at random from a consignment of fuses from the factory. Find the probability that

a) The fuse is faulty,

b) The fuse is faulty or it comes from Y

c) The fuse comes from X given that it is faulty **(2+ 2+ 3 = 7 Marks)**